Coding & Solution

Date	12 July 2026
Team ID	-
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	SK-P-BibiAyesha/skill-wallet-google-cloud-generative-ai: Voice-Based Concept Understanding Analyser using Faster Whisper and Sentence-BERT
Programming Language(s)	Python, HTML, CSS

Framework(s) Used	Streamlit
Key Features Implemented	Audio Upload, Speech-to-Text, Semantic Similarity, Keyword Matching, Audio Analysis, Score Generation, AI Feedback, PDF Report Generation, Dashboard, Analysis History
Pending / Incomplete Features	User Authentication, Cloud Deployment, Live Voice Recording, Multi-language Support (optional if not implemented)
Setup / Run Instructions	Install requirements (pip install -r requirements.txt) → Run streamlit run app.py

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	yes
2	Meaningful variable and function names are used	yes
3	Code includes comments / documentation where necessary	yes
4	Error handling is implemented for critical operations	yes
5	The application runs without critical errors	yes
6	Code is committed to a version control repository	yes

Additional Notes / Comments:

The project follows good software engineering practices using a modular architecture and reusable components. AI-powered speech recognition, semantic analysis, keyword matching, scoring, and automated report generation work together to evaluate conceptual understanding efficiently. The codebase is version-controlled using GitHub and can be extended with authentication, cloud deployment, and multilingual support in future releases.